Solving Quadratics — Incomplete Root Sets and Sign Errors Answer Key

Algebra 1

Quick Answers

1. -7, 7
2. 0, 3
3. 121
4. $1 + \frac{\sqrt{10}}{2}$
5. -2, 8
6. $-\frac{3}{2}$, 4
7. 0, $\frac{3}{2}$
8. —

Common wrong answers

3. *If they got -23: lost the negative sign on c and subtracted 72 instead of adding it*
4. *If they got 0.58: wrote the numerator as minus four because b is already negative, instead of negating b to plus four*
7. *If they got 288: added four a c instead of subtracting it, turning a perfect square into a positive discriminant*

1. Solve $x^2 = 49$; is $x = 7$ complete?

Step 1: rewrite so one side is zero — $x^2 - 49 = 0$ — because every method in this unit reads a quadratic in that form.
Step 2: this is a difference of squares: $(x-7)(x+7) = 0$. The zero-product property gives $x - 7 = 0$ or $x + 7 = 0$.
Step 3: both factors are legitimate, so $\boxed{x = -7 \text{ or } x = 7}$. Toby's set is *incomplete*: he took the principal square root of 49 and stopped, but $(-7)^2 = 49$ as well.
Teaching point: the square-root step is where roots go missing. Writing $x = \pm\sqrt{49}$ keeps both.

2. Find and fix: Jae divides $x^2 = 3x$ by x.

Step 1: name the assumption. Dividing both sides by x is only legal when $x \neq 0$, and nothing in the equation says that. Jae divided away the very solution $x = 0$.
Step 2: solve correctly by moving everything to one side and factoring: $x^2 - 3x = 0$, so $x(x-3) = 0$.
Step 3: the zero-product property gives $x = 0$ or $x - 3 = 0$, so $\boxed{x = 0 \text{ or } x = 3}$.
Teaching point: never divide both sides by an expression containing the variable. Factor instead — factoring keeps every root, division destroys one.

3. Find and fix: Rio's discriminant for $3x^2 - 7x - 6 = 0$.

Step 1: list the coefficients exactly as the equation writes them, signs included: $a = 3$, $b = -7$, $c = -6$. Rio substituted c as $+6$ — he took the number and dropped its sign.
Step 2: with the correct c, the term $-4ac$ becomes $-4(3)(-6) = +72$: a negative c makes that term *add*, which is exactly the case Rio's shortcut cannot survive.
Step 3: $b^2 - 4ac = (-7)^2 + 72 = 49 + 72 = \boxed{121}$.
Step 4: $121 > 0$ and is a perfect square, so the equation has two distinct rational solutions — the opposite of Rio's conclusion. (For the record they are $x = 3$ and $x = -\frac{2}{3}$.)
Watch for: -23, which is $49 - 72$. A negative discriminant is the signature of this slip, and it wrongly reports "no real solutions" on an equation that factors.

4. Find and fix: Mia's numerator on $2x^2 - 4x - 3 = 0$.

Step 1: state the rule. The formula's numerator opens with $-b$, which means *the opposite of* b — not "a minus sign in front of the number you see". Here $b = -4$, so $-b = +4$.
Step 2: with $a = 2$, $b = -4$, $c = -3$: $b^2 - 4ac = 16 + 24 = 40$, and $\sqrt{40} = 2\sqrt{10}$.
Step 3: $x = \dfrac{4 \pm 2\sqrt{10}}{4} = 1 \pm \dfrac{\sqrt{10}}{2}$, so the larger root is $\boxed{x = 1 + \dfrac{\sqrt{10}}{2}}$, about 2.58.
Watch for: 0.58. That is Mia's value — her numerator started at -4 instead of $+4$, so her root is exactly 2 less than the true one. A quick substitution check catches it: her value does not satisfy the original equation.

5. Find and fix: Sam square-roots $(x - 3)^2 = 25$.

Step 1: the lost line is the square-root step. $\sqrt{(x-3)^2} = |x - 3|$, so the correct line reads $x - 3 = \pm 5$, not $x - 3 = 5$.
Step 2: split into the two branches: $x - 3 = 5$ gives $x = 8$; $x - 3 = -5$ gives $x = -2$. Sam kept the first branch and dropped the second.
Step 3: the complete solution set is $\boxed{x = -2 \text{ or } x = 8}$.
Step 4: check both in the original: $(8 - 3)^2 = 25$ ✓ and $(-2 - 3)^2 = 25$ ✓.
Teaching point: expanding instead would give $x^2 - 6x - 16 = 0$ and factor to $(x - 8)(x + 2)$ — the same two roots. Two methods, one complete set.

6. Find and fix: Kai reads $(2x + 3)(x - 4) = 0$ off the page.

Step 1: first mistake — *sign.* Setting a factor to zero flips the sign of the constant: $x - 4 = 0$ gives $x = +4$, not -4, and $2x + 3 = 0$ gives a negative x, not $+3$. Kai copied the numbers and kept their printed signs.
Step 2: second mistake — *an unfinished step.* The factor $2x + 3$ has a coefficient, so solving it takes two moves: $2x = -3$, then divide by 2.
Step 3: the solutions are $2x + 3 = 0 \Rightarrow x = -\frac{3}{2}$ and $x - 4 = 0 \Rightarrow x = 4$, so $\boxed{x = -\frac{3}{2} \text{ or } x = 4}$.
Teaching point: Kai's answers are each the negative of a true root or an unfinished one. Substituting $x = 3$ gives $(9)(-1) = -9 \neq 0$, which is a two-second refutation.

7. Find and fix: is $4x^2 - 12x + 9 = 0$ really two-rooted?

(a) Step 1: read the coefficients with their signs: $a = 4$, $b = -12$, $c = 9$. The discriminant decides how many real solutions exist before any root is computed.
Step 2: $b^2 - 4ac = (-12)^2 - 4(4)(9) = 144 - 144$, so $\boxed{\Delta = 0}$.
(b) Step 3: a zero discriminant means the $\pm$ contributes nothing, so there is exactly *one* distinct real solution (a double root): $x = \dfrac{-b}{2a} = \dfrac{12}{8}$, giving $\boxed{x = \frac{3}{2}}$.
Step 4: confirm by factoring — $4x^2 - 12x + 9 = (2x - 3)^2$, a perfect square, so the only zero is where $2x - 3 = 0$.
Step 5: the student's rule is wrong twice over. Roots do not come in positive/negative pairs (that only happens when $b = 0$), and a quadratic has two, one, or no real solutions depending on the sign of the discriminant.
Watch for: 288, from $(-12)^2 + 4(4)(9)$ — adding $4ac$ instead of subtracting it. That turns a perfect square into a two-root equation and hides the double root entirely.

8. Explain: why $\pm$ is not optional.

Open response — flagged for manual review. Model answer below; accept any equivalent reasoning.
(a) Squaring destroys sign information: a number and its opposite have the same square, so an equation of the form $u^2 = k$ (with $k > 0$) has two solutions, $u = \sqrt{k}$ and $u = -\sqrt{k}$. Writing only $u = \sqrt{k}$ keeps the branch you happened to think of and silently discards the other. The honest step is $u = \pm\sqrt{k}$, or equivalently $|u| = \sqrt{k}$.
(b) The same $\pm$ sits in the numerator of $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. It is what the square-root step in completing the square left behind, and it is what produces two roots: one from the plus branch, one from the minus branch. When $b^2 - 4ac = 0$ the two branches land on the same number, which is why a double root is one solution, not two.
(c) Model habit: "Before I hand it in, I count my solutions against the discriminant, and I substitute every root back into the original equation. If I have fewer roots than the discriminant predicts, I go back to the line where I took a square root or divided by something containing x."
What a full-credit answer must contain: the observation that two numbers share one square, the identification of $\pm$ in the formula as the source of the second root, the discriminant-zero case named as the exception, and a habit that is a check rather than a restatement of the rule.